AMA515A Forecasting and applied time series analysis

Chapter 4 : Statistical Analysis in the Time Domain

4.1 Variance stability transformations

A common phenomenon in non-stationary time series data is that the variability is related to the average level.

Consider a r.v. X whose distribution is indexed by a parameter θ such that E(X)= θ and Var(X)=v(X)= $v^2(\theta)$. $= [v(\theta)]^2$

Consider the transformation Y=h(X) s.t. v(Y)= σ^2, independent of θ. Then

$$h(X) \approx h(\theta) + (X-\theta)h'(\theta)$$ Taylor series expansion about θ.

$$\Rightarrow \{h(X)-h(\theta)\}^2 \approx (X-\theta)^2\{h'(\theta)\}^2$$
$$\Rightarrow v(Y) \approx v(X)\{h'(\theta)\}^2$$
$$\sigma^2 = v^2(\theta)\{h'(\theta)\}^2 \qquad (4.1)$$
$\sigma = v(\theta)h'(\theta)$

(1) Thus assume $v^2(\theta)$ known, we can find h(·).
Example : Poisson data

(4.1) implies $\{h'(\theta)\}^2 = \dfrac{\sigma^2}{v^2(\theta)}$ $h'(\theta) = \dfrac{\sigma}{v(\theta)} \Rightarrow \int h'(\theta)d\theta = \int \dfrac{\sigma}{v(\theta)}d\theta$

$$\Rightarrow h(\theta) = \int \frac{\sigma}{v(\theta)}d\theta$$

Thus $v^2(\theta)=\theta$ implies $v(\theta)=\theta^{\frac{1}{2}}$, whence $h(\theta) = \int \dfrac{\sigma}{\theta^{0.5}} \propto \theta^{\frac{1}{2}}$, i.e., square root transformation stabilizes variance of Poisson data. $d\theta$

(2) v(θ) unknown, seek h(.) on empirical ground.
A general class of power transformation is $\sigma\int \theta^{-\frac{1}{2}}d\theta$

$$h(\theta) = \begin{cases} (\theta^\lambda - 1)/\lambda & : \lambda \neq 0 \\ \log\theta : & \lambda = 0 \end{cases}$$ $\ln\theta$ $= 2\sigma\theta^{\frac{1}{2}} + C \propto \theta^{\frac{1}{2}}$

$$h'(\theta) = \theta^{\lambda-1}$$

Reconsider (4.1) $\sigma = v(\theta).h'(\theta)$, which implies

$h(\theta) = \dfrac{\theta^\lambda - 1}{\lambda}$ $h'(\theta) = \dfrac{\lambda\theta^{\lambda-1}}{\lambda} = \theta^{\lambda-1}$

$\log(v(\theta)) = \log\sigma + (1-\lambda)\log\bar{x}$,

and hence a practical way to estimate λ is by using a plot of

$\log(s) = \log\sigma + (1-\lambda)\log\bar{x}$.

Namely we divide the time series data into a number of blocks and find the sample standard deviation (s.d.) and sample mean of each block. We then plot the sample standard deviation against the sample mean and find the estimate of λ from the slope of the plot.

4.2 Partial Autocorrelation

Correlogram is a crucial tool in recognition of plausible models for time series data. But because of sampling fluctuations, cut-off correlogram is easier to recognize than exponential decay.

The notion of partial autocorrelations is designed to help the model identification procedure by exploiting this.

Let $\rho = (\rho_1,\ldots,\rho_k)^T$, vector of 1st k autocorrelations of an AR(k) process.
Let R_k be k x k matrix with (i,j) element $\rho_{|i-j|}$. Let $\alpha = (\alpha_1,\ldots,\alpha_k)^T$ be the vector of parameters . Recall the Yule-Walker equations

$$\rho = \mathbf{R_k}\alpha, \text{ thus } \quad \alpha = \mathbf{R_k}^{-1}\rho$$

and so $\alpha_k = |\mathbf{R_k}|^{-1}|\mathbf{R_k}^*|$, where $\mathbf{R_k}^*$ denotes $\mathbf{R_k}$ with its final column replaced by $(\rho_1,\ldots,\rho_k)^T$.

Now if the process is actually AR(k-1), $\alpha_k = 0$. Taking $\hat{\mathbf{R}}_\mathbf{k}$ *and* $\hat{\mathbf{R}}_\mathbf{k}^*$ to be empirical version of $\mathbf{R_k}$ and $\mathbf{R_k}^*$, define the partial correlogram to be the set of coefficients

$$\hat{\phi}_k = |\hat{\mathbf{R}}_\mathbf{k}^*| / |\hat{\mathbf{R}}_\mathbf{k}|, \quad k=1,2,\ldots$$

So, for an AR(p) process, we expect $\hat{\phi}_k \approx 0$, for all k>p. Also, $\hat{\phi}_k$ has asymptotic variance n^{-1}, so use $\pm 2/\sqrt{n}$ to look for significantly non-zero $\hat{\phi}_k$, just as for ordinary correlogram.

4.3 Model identification

Within the general framework of (possibly seasonal) ARIMA(p,d,q) processes statistical analysis proceeds in three stages:

(i) Identify plausible choices for p,d,q
(ii) Estimate parameters
(iii) Check that residuals are consistent with white noise.

$\rho(k)$ ψ_{kk}

Two major tools in (i) are the correlogram and the partial correlogram defined above. Other tools include time plots, differencing, experience and common sense.

- Time plots: essential preliminary to any analysis. Outliers? Non-stationarity? Variance instability?
- If time series is non-stationary, then ACF typically dies down very slowly from 1
- Differencing: Can eliminate both deterministic and stochastic trends in data. Over differencing tends to increase variability. Under differencing tends to produce a correlogram which decays slowly or not at all.
- Correlogram: Can resolve doubtful cases of stationarity. More positively, seek to relate correlogram to one of the various distinctive theoretical ACF of sub-classes of ARMA processes(exponential decay, cut-off…). Partial correlogram also help. Conventional error is $\pm 2/\sqrt{n}$

i.e. ARIMA(p,d,q)

4.3.1 Behavior of the acf and pacf for non-seasonal ARIMA models

Model	Autocorrelation function	Partial autocorr. function
Moving average of order q	Cuts off after lag q	Dies down exponentially and/or sinusoidally
Autoregressive of order p	Dies down exponentially and/or sinusoidally	Cuts off after lag p
Mixed model of order p,q	Dies down exponentially and/or sinusoidally	Dies down exponentially and/or sinusoidally

MA

AR

ARMA

⇒ die down slowly

4.3.2 Characteristics and equations for the most common Box-Jenkins models [ARIMA(p,d,q)(P,D,Q)$_L$]

Note : both seasonal and nonseasonal models are included

Acf	Pacf	model
Cuts off after lag 1 or 2; no significant spikes at the seasonal lags	Dies down	Non-seasonal MA (q=1 or 2) e.g. $X_t = \mu - \theta_1 a_{t-1} + a_t$, where a_t is NID(0,σ^2).
Cuts off after seasonal lag L; no significant spikes at the nonseasonal lags	Dies down	Seasonal MA e.g. $X_t = \mu - \theta_{1L} a_{t-L} + a_t$
Cuts off after seasonal lag L; significant spikes also exist at nonseasonal lags 1 or 2	Dies down	Nonseasonal-seasonal MA (q=1 or 2; Q=1) e.g. $X_t = \mu - \theta_1 a_{t-1} - \theta_{1L} a_{t-L} + \theta_1 \theta_{1L} a_{t-L-1} + a_t$
Dies down	Cuts off after lags 1 or 2; no significant spikes at the seasonal level	Nonseasonal autoregressive (p=1 or 2) e.g. $X_t = \mu + \phi_1 X_{t-1} + a_t$
Dies down	Cuts off after seasonal lag L; no significant spikes at the nonseasonal lags	Seasonal autoregressive (P=1) $X_t = \mu + \phi_{1L} X_{t-L} + a_t$
Dies down	Cuts off after seasonal lag L; significant spikes also exist at nonseasonal lags 1 or 2	Nonseasonal-seasonal autoregressive (p=1 or 2; P=1) e.g. $X_t = \mu + \phi_1 X_{t-1} + \phi_{1L} X_{t-L} + \phi_1 \phi_{1L} X_{t-L-1} + a_t$
Dies down	Dies down	Mixed (autoregressive-moving average) Nonseasonal: e.g. $X_t = \mu + \phi_1 X_{t-1} + \theta_1 a_{t-1} + a_t$ seasonal: e.g. $X_t = \mu + \phi_{1L} X_{t-L} - \theta_{1L} a_{t-L} + a_t$

4.3.3 Additional approach to model identification

Many approaches to model identification have been proposed since Box and Jenkins' work. A popular one is the Akaike's Information Criterion (AIC, 1973). AIC says to select the model that minimizes

$$\text{AIC} = -2\log(\text{maximum likelihood}) + 2k,$$

where k = p+q+1 if the ARMA model contains a constant term, and k= p+q otherwise. Maximum likelihood estimation is discussed in the next section. The addition of the term 2(p+q+1) or 2(p+q) serves as penalty function to avoid choosing models with many parameters. However it was shown that AIC tends to select models with more parameters in similar models (Shibata, 1976), thus many other information criteria have appeared in the literature. Two well-known ones are:

(a) AICC proposed by Hurvich and Tsai (1979)

$$\text{AICC} = AIC + \frac{2(k+1)(k+2)}{n-k-2}, \text{ and}$$

(b) Bayesian information criterioin (BIC, 1978)

$$\text{BIC} = -2\log(\text{maximum likelihood}) + k\log(n)$$

Both criteria will tend to selects models with smaller k.

4.4 Parameter estimation

Having identify an ARMA(p,q) process for the suitably differenced data, we need to estimate p and q parameters; we use least squares, i.e. we minimize sum of squared estimated residuals. Alternatively we can use the maximum likelihood approach.

Definition: All ARMA process involve an embedded white noise sequence $\{Z_t\}$. The sequence of values $\{\hat{Z}_t\}$ recovered from the data $\{\hat{x}_t\}$ and the fitted parameters $\hat{\alpha}_i, \hat{\beta}_j$ are called the residuals. If the model is suitable(reasonable), the residuals should approximately be a realization of white noise(uncorrelated, zero mean, constant variance).

4.4.1 Autoregressive processes

$$X_t = \sum_{i=1}^{p} \alpha_i X_{t-i} + Z_t \quad \text{iff} \quad Z_t = X_t - \sum_{i=1}^{p} \alpha_i X_{t-i}$$

Estimate $\hat{\alpha}_i$ s.t. $\sum_{t=p+1}^{n}(x_t - \sum_{i=1}^{p}\alpha_i x_{t-i})^2$ is a minimum. with p equations in p unknowns.

N.B. Any non-zero mean can be eliminated at the identification stage.

L.S. least squares

The L.S. solution is given by the Yule-Walker equation, $\alpha = \mathbf{R}^{-1}\rho$,

where $\rho = (\rho_1,\ldots,\rho_p)^T$, the vector of the 1st p sample autocorrelation coefficients, and R is a p x p matrix with $(i,j)^{th}$ elements $\rho_{|i-j|}$. $\rho(k) = \rho(-k)$

4.4.2 Moving average processes (MA(q))

$$X_t = Z_t + \sum_{j=1}^{q}\theta_j Z_{t-j} \quad \text{iff} \quad Z_t = X_t - \sum_{j=1}^{q}\theta_j Z_{t-j}$$ with q unknowns.

Residuals must be calculated recursively. Since the expectation $E[Z_t]= 0$, for all t, take unobservable $Z_0, Z_{-1},\ldots$ zero and calculate

$$Z_1^* = x_1,\ Z_2^* = x_2 - \theta_1 Z_1^*,\ Z_3^* = x_3 - \theta_1 Z_2^* - \theta_2 Z_1^*,\ Z_{q+1}^* = x_{q+1} - \sum_{j=1}^{q}\theta_j Z_{q+1-j},\ldots$$

Then minimize $\sum_{t=1}^{n} Z_t^{*2}$ numerically.

4.4.3. Mixed processes (ARMA(p,q))

$$X_t = \sum_{i=1}^{p}\alpha_i X_{t-i} + Z_t + \sum_{j=1}^{q}\beta_j Z_{t-j} \quad \text{iff} \quad X_t - \sum_{i=1}^{p}\alpha_i X_{t-i} - \sum_{j=1}^{q}\beta_j Z_{t-j} = Z_t$$

Then calculate Z_t^* recursively as above.

Ex. ARMA(1,1)

$$X_t - \mu = \alpha_1(X_{t-1} - \mu) + Z_t + \beta_1 Z_{t-1}$$

Given N observations $x_1, x_2,\ldots, x_N$, we guess values for μ, α_1, β_1. Set $Z_0^*=0$ and $x_0=\mu$, and then calculate the residuals recursively by

$$Z_1^* = x_1 - \mu,\ Z_2^* = x_2 - \mu - \alpha_1(x_1 - \mu) - \beta_1 Z_1^*,\ \ldots..$$

$$Z_N^* = (x_N - \mu) - \alpha_1(x_{N-1} - \mu) - \beta_1 Z_{N-1}^*$$

The residual sum of squares $\sum_{t=1}^{N} Z_t^{*2}$ may then be calculated. Then other values of α_1, β_1 may be tried until a minimum is found.

Note: A more general and rigorous theory on the estimation of parameters in ARIMA models is based on maximum likelihood. For the interested students, you may

QA280 William W.S. Wei

refer to Chapter 3 of Harvey (Time Series Models, 1993) or Ch. 7 of Wei (Time series analysis : univariate and multivariate methods, 1993). A brief account is shown below.

4.5 Maximum likelihood estimation

We begin with the non-time series case. Suppose that we have a sample of independent observations X_t, t=1,2,…,N each with pdf f(X_t). Then the joint density function is

$$f(X_1, X_2, ..., X_N) = \prod_{t=1}^{N} f(X_t). \qquad (4.2)$$

Let X_t contain an unknown parameter θ (it can also be a vector). The likelihood function is the joint density (4.2), but is regarded as a function of the parameter θ, and as such is written

$$L(\theta) = \prod_{t=1}^{N} f(X_t; \theta).$$

The maximum likelihood estimates (MLEs) are naturally found by maximizing L(θ) with respect to θ. A usual trick is to maximize log (L(θ)) instead of L(θ). It's because the log function is a concave function. The problem with time series is that the observations are not independent, so that (4.2) is not applicable. If, however, we denote all observations up to and including X_t by XX_t (which is a vector in general) we can write

given

$$f(X_1, X_2, ..., X_N) = f(X_N \mid XX_{N-1}) f(X_1, X_2, ...X_{N-1})$$
$$= f(X_N \mid XX_{N-1}) f(X_{N-1} \mid XX_{N-2}) f(X_1, X_2, ..., X_{N-2}),$$

and so on until

given ↑ given ↑

$$f(X_1, X_2, ..., X_N) = \prod_{t=1}^{N} f(X_t \mid XX_{t-1})$$

parameters of time series

this involves $\alpha_1, \alpha_2, \ldots \alpha_p$ and $\beta_1, \beta_2, \ldots \beta_q$

where $f(X_t \mid XX_{t-1})$ is the conditional distribution of X_t given all observations prior to t. So the likelihood L(θ) can be written as the product of these conditional distributions.

Example 1: AR2 model

Let the model be $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + a_t$, $\quad E(X_t) = \phi_1 E(X_{t-1}) + \phi_2 E(X_{t-2})$

where $\{a_t\}$ is a white noise series, or a series of normally distributed, independent random variables with zero mean and variance σ^2.

X_{t-2}

Then the mean of the conditional distribution (which is normal) $f(X_t \mid X_1, X_2, ..., X_{t-1})$ is $\phi_1 X_{t-1} + \phi_2 X_{t-2}$, and the variance of the conditional distribution is σ^2, and therefore

$E(X_t) = \phi_1 X_{t-1} + \phi_2 X_{t-2}$ Since X_{t-1} and X_{t-2} are given values

$Var(X_t) = \phi_1^2 Var(X_{t-1}) + \phi_2^2 Var(X_{t-2}) + Var(a_t)$

$= 0 + 0 + \sigma^2$

$$f(X_t \mid X_1, X_2, \ldots, X_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[\frac{-(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2}{2\sigma^2}\right].$$

Problems arise for t=1 and t=2 because $f(X_1)$ and $f(X_2|X_1)$ are not known. To get around this, we assume that the first two observations X_1, X_2 are fixed and define the conditional likelihood as

$$L(\phi_1, \phi_2, \ldots) \leftarrow L(\theta) = \prod_{t=3}^{N} f(X_t \mid XX_{t-1}).$$

N.B. From 3 to N, there we have N-3+1 = N-2 terms

The conditional log-likelihood is therefore

$$\ln L(\phi_1, \phi_2) = \log L(\phi_1, \phi_2) = l(\phi_1, \phi_2) = -\frac{(N-2)}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=3}^{N}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2.$$

If σ^2 is known, maximizing the conditional likelihood is equivalent to minimizing

$$\sum_{t=3}^{N}(X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2})^2. \qquad (4.3)$$

That means the conditional maximum likelihood estimator is equivalent to the conditional least squares estimator. We stress that this is not always the case. In general we have to use nonlinear optimization techniques, such as the Newton's method. See the next example on MA1 model.

If σ^2 is unknown, it can be shown from direct differentiation that the maximum likelihood estimator of σ^2 is the minimum sum of squares from (4.3) divided by the number of terms included in the sum.

Linear model $MSE = \frac{SSE}{n-k-1} = \hat{\sigma}^2$

k is the number of regressors in the model.

Note: Newton's method

Let f(x)=0 be a nonlinear algebraic equation in one unknown. Then a well-known algorithm, known as Newton's method, for finding the roots of the equation is given by the following iterative scheme.

$$x_{n+1} = x_n - f(x_n)/f'(x_n).$$

Illustration: Find a root of $x^3-2x-5=0$ near 2 to 4 decimal places.
Solution: The Newton's scheme is

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

Let $x_0=2$, then

$$x_1 = 2-(-1)/10 = 2.1,$$
$$x_2 = 2.1-\frac{0.0061}{11.23} = 2.09457,$$
$$x_3 = 2.09455$$

Thus the root is 2.0946.

Example 2 : MA1 model

We have a time series data set $\{x_1, x_2, \ldots, x_n\}$ and we believe the model is MA1, i.e.,

$$X_t = \theta a_{t-1} + a_t,$$

where $\{a_t\}$ is a white noise series as in the above example, i.e., normal variables with mean zero and variance σ^2.

We want to estimate $\theta \,\&\, \sigma^2$. Using similar argument as Example1, the conditional likelihood is

$\prod_{t=2}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x_t-\theta a_{t-1})^2}{2\sigma^2}\right\}$, and hence the conditional log-likelihood is, apart from a constant,

$l(\theta,\sigma^2) = l = -(n-1)\log(\sigma^2)/2 - \sum_{t=2}^{n} \frac{(x_t-\theta a_t)^2}{2\sigma^2}$. To simplify the algebra, let $\sigma^2 = s$. Then

$$l = -(n-1)\log(s)/2 - \sum_{t=2}^{n} \frac{(x_t-\theta a_t)^2}{2s} \qquad (4.4)$$

The maximum likelihood procedure is to solve the two equations $\frac{\partial l}{\partial \theta} = 0$, & $\frac{\partial l}{\partial s} = 0$.

It turns out we basically only need to solve $\frac{\partial l}{\partial \theta} = 0$. Note that this is a nonlinear equation in θ. We can use Newton's method. That means we need to deal with

$$\theta_{n+1} = \theta_n - \left.\frac{\partial^2 l}{\partial^2 \theta}\right|_{\theta_n}^{-1} \left.\frac{\partial l}{\partial \theta}\right|_{\theta_n}. \qquad (4.5)$$

$$\theta_{n+1} = \theta_n - \frac{\left.\frac{\partial l}{\partial \theta}\right|_{\theta_n}}{\left.\frac{\partial^2 l}{\partial \theta^2}\right|_{\theta_n}}$$

To carry out the iterations in (4.5), we have to calculate the 1st and 2nd derivatives in the equation.

Now $a_t = x_t - \theta a_{t-1}$. This is an important difference from the AR case, because a_t is a function of θ. Thus

$$\frac{\partial a_t}{\partial \theta} = -\theta \frac{\partial a_{t-1}}{\partial \theta} - a_{t-1}, \quad \frac{\partial^2 a_t}{\partial^2 \theta} = -2\frac{\partial a_{t-1}}{\partial \theta} - \theta \frac{\partial^2 a_{t-1}}{\partial \theta^2}$$

From (4.4), $\partial l / \partial \theta = -\frac{1}{2}\sum_{t=2}^{n} -2(\frac{x_t - \theta a_t}{s})\left[\theta \frac{\partial a_t}{\partial \theta} + a_t\right]$,

and $$\partial^2 l / \partial \theta^2 = \frac{1}{s}\sum_{t=2}^{n} -\left(\theta \frac{\partial a_t}{\partial \theta} + a_t\right)^2 + (x_t - \theta a_t)(2\frac{\partial a_t}{\partial \theta} + \theta \frac{\partial^2 a_t}{\partial \theta^2}).$$

Note that we have to make assumptions on the initial values, i.e, θ_0, $a_0, \frac{\partial a_0}{\partial \theta}, \& \frac{\partial^2 a_0}{\partial \theta^2}$.

So the MA case is considerably more complicated than the AR case.

Finally, differentiating (4.4) with respect to s gives us

$$\frac{\partial l}{\partial s} = \sum_{t=2}^{n} 1/s - \frac{(x_t - \theta a_t)^2}{s^2} = 0, \text{ which implies } s = \sum_{t=2}^{n} (x_t - \hat{\theta} a_t)^2 /(n-1).$$

Here $\hat{\theta}$ is the estimate of θ obtained from (4.5).

4.6 Diagnostic checking (Residual analysis)

Check that the estimated residual series $\{\hat{Z}_t\}$ is consistent with white noise. Same techniques are relevant here. General guide, use

(i) time plot of $\{\hat{Z}_t\}$

(ii) correlogram of residuals

(iii) cumulative periodogram test (a frequency domain technique)

(iv) trial over-fitting, i.e., examine residual sum of squares for chosen model and suitably chose extensions. Usually calculate $100(1-\frac{\text{var}(\hat{Z})}{\text{var}(X)})$, i.e.,% of variance explained by model.

(v) Box-Pierce statistics and Ljung-Box statistics (χ^2 test)

B-P : $n\sum_{k=1}^{m}\hat{r}^2(k)$ follows χ^2_{m-p-q} dn. asymptotically. Namely, it works for large samples.

L-B : $n(n+2)\sum_{k=1}^{m}\hat{r}^2(k)/(n-k)$ follows χ^2_{m-p-q} dn., which works better for small sample.

Proposition:
Let e_t be a white noise sequence, namely, i.i.d. random variables with zero mean and constant variance 1. Consider the lag-1 autocorrelation coefficient of e_t, ρ_1, which is given by:

$$\rho_1 = \frac{\sum_{t=1}^{n-1} e_t e_{t+1}}{\sum_{t=1}^{n} e^2{}_t}.$$

Then $\mathrm{var}(\rho_1) \doteq 1/n$.

Proof: Let $X=\sum_{t=1}^{n-1} e_t e_{t+1}$, $Y=\sum_{t=1}^{n} e^2{}_t$, then $\rho_1 = \frac{X}{Y}$. Observe

$$E(X) = \mu_x = 0, \quad E(Y) = \mu_y = n.$$

Let f(X,Y)= X/Y. Consider the Taylor expansion of f(X,Y) about the point

$$(\mu_x, \mu_y) = (0, n).$$

Then $f(X,Y) \doteq f(\mu_x,\mu_y) + \left(\frac{\partial f}{\partial X}\right)_{(\mu_x,\mu_y)} \cdot (X-\mu_x) + \left(\frac{\partial f}{\partial Y}\right)_{(\mu_x,\mu_y)} \cdot (Y-\mu_y)$.

$= 0 + X/n + 0$.

Thus $\mathrm{var}(X/Y) \doteq \mathrm{var}(X/n)$. But $\mathrm{var}(X) = \mathrm{var}(\sum_{t=1}^{n-1} e_t e_{t+1}) = E[(\sum_{1}^{n-1} e_t e_{t+1})^2] - 0$

$$= E(\sum_{t=1}^{n-1} e_t{}^2 e_{t+1}{}^2) + E(\sum_{t}\sum_{t'} e_t e_{t+1} e_{t'} e_{t'+1}).$$

The second term above vanishes as e_t is white noise and $t \neq t'$, and so

$\mathrm{var}(X) = E(\sum_{t=1}^{n-1} e_t{}^2 e_{t+1}{}^2) = n-1$.

Then we have $\mathrm{var}(\rho_1) = (n-1)/n^2 \doteq 1/n$.

Example on model fitting:

Data-- US quarterly unemployment data, 1948-1978 (121 observations, from Cryer 1982)

unemploy	ar2-residuals
3.73	-0.13484
3.67	-0.07855
3.77	0.00441
3.83	-0.13106
4.67	0.68093
5.87	0.60922
6.70	0.11761
6.97	-0.12018
6.40	-0.56678
5.57	-0.32987
4.63	-0.34854
4.23	0.17382
3.50	-0.54888
3.10	-0.07204
3.17	0.14311
3.37	-0.02872
3.07	-0.59494
2.97	-0.09540
3.27	0.16376
2.83	-0.81014
2.70	-0.05644
2.57	-0.27236
2.73	0.00521
3.70	0.64015
5.27	0.80116
5.80	-0.48262
5.97	-0.10928
5.33	-0.66672
4.73	-0.15620
4.40	0.03019
4.10	-0.14858
4.23	0.23305
4.03	-0.36678
4.20	0.20072
4.13	-0.26591
4.13	-0.04506
3.93	-0.29101
4.10	0.19116
4.23	-0.07547
4.93	0.53322
6.30	0.89597
7.37	0.28712

(52)

7.33	-0.52368
6.37	-0.71888
5.83	0.21326
5.10	-0.30405
5.27	0.65089
5.60	0.23637
5.13	-0.63712
5.23	0.41309
5.53	0.24849
6.27	0.58588
6.80	0.15779
7.00	0.01630
6.77	-0.17796
6.20	-0.25769
5.63	-0.08899
5.53	0.32653
5.57	0.14845
5.53	-0.01962
5.77	0.30907
5.73	-0.13179
5.50	-0.14182
5.57	0.26092
5.47	-0.09932
5.20	-0.16728
5.00	-0.01150
5.00	0.12343
4.90	-0.10785
4.67	-0.18177
4.37	-0.18842
4.10	-0.14114
3.87	-0.14664
3.80	-0.03488
3.77	-0.10660
3.70	-0.17573
3.77	-0.01616
3.83	-0.11137
3.83	-0.15907
3.93	-0.01968
3.73	-0.37577
3.57	-0.15796
3.53	-0.07951
3.43	-0.22210
3.37	-0.15228
3.43	-0.06427
3.60	-0.02730
3.60	-0.25326

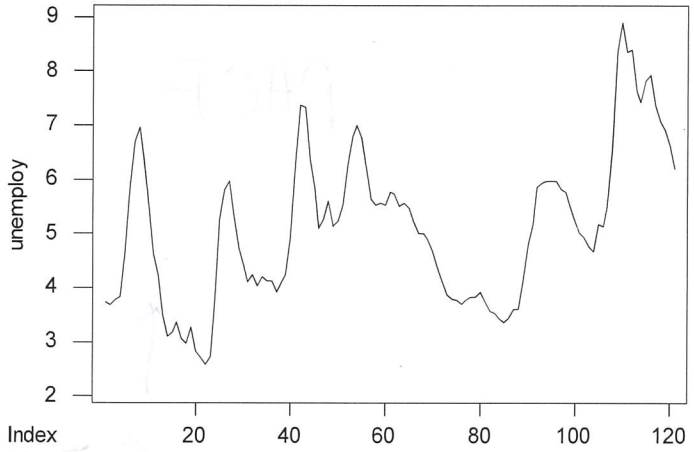
(98)

4.17	0.42833
4.80	0.16866
5.17	-0.07051
5.87	0.46552
5.93	-0.32418
5.97	0.08166
5.97	0.05861
5.97	0.08487
5.83	-0.05513
5.77	0.10338
5.53	-0.13486
5.27	-0.05965
5.03	-0.05137
4.93	0.05256
4.77	-0.10890
4.67	-0.02481
5.17	0.52625
5.13	-0.35981
5.50	0.40083
6.57	0.86707
8.37	1.23985
8.90	-0.33728
8.37	-0.51297
8.40	0.69217
7.63	-0.47255
7.43	0.54898
7.83	0.75570
7.93	0.10009
7.37	-0.35343
7.07	0.28628
6.90	0.21694
6.63	0.01535
6.20	-0.10481

120th → 6.63

121th → 6.20

(131)




```
MTB > acf c1
```

Autocorrelation Function: unemploy

ACF

```
ACF of unemploy

          -1.0 -0.8 -0.6 -0.4 -0.2  0.0  0.2  0.4  0.6  0.8  1.0
            +----+----+----+----+----+----+----+----+----+----+----+
  1    0.936                           XXXXXXXXXXXXXXXXXXXXXXXXX
  2    0.802                           XXXXXXXXXXXXXXXXXXXXX
  3    0.646                           XXXXXXXXXXXXXXXXX
  4    0.502                           XXXXXXXXXXXXXX
  5    0.392                           XXXXXXXXXXX
  6    0.317                           XXXXXXXXX
  7    0.263                           XXXXXXXX
  8    0.223                           XXXXXXX
  9    0.197                           XXXXXX
 10    0.172                           XXXXX
 11    0.144                           XXXXX
 12    0.115                           XXXX
 13    0.098                           XXX
 14    0.095                           XXX
 15    0.109                           XXXX
 16    0.127                           XXXX
 17    0.126                           XXXX
 18    0.099                           XXX
 19    0.046                           XX
 20   -0.025                         XX
 21   -0.102                       XXXX
 22   -0.171                      XXXXX
 23   -0.229                    XXXXXXX
 24   -0.272                  XXXXXXXX
 25   -0.293                  XXXXXXXX
 26   -0.305                XXXXXXXXXX
```

```
 27  -0.319                    XXXXXXXXX
 28  -0.340                   XXXXXXXXXX
 29  -0.364                   XXXXXXXXXX
 30  -0.384                  XXXXXXXXXXX
```

```
MTB > pacf 20 c1
```

Partial Autocorrelation Function: unemploy

PACF

```
PACF of unemploy

          -1.0 -0.8 -0.6 -0.4 -0.2  0.0  0.2  0.4  0.6  0.8  1.0
            +----+----+----+----+----+----+----+----+----+----+----+
  1   0.936                             XXXXXXXXXXXXXXXXXXXXXXXXX
  2  -0.605           XXXXXXXXXXXXXXXX
  3   0.098                             XXX
  4   0.052                             XX
  5   0.120                             XXXX
  6  -0.033                            XX
  7  -0.049                            XX
  8   0.022                             XX
  9   0.098                             XXX
 10  -0.138                          XXXX
 11   0.018                             X
 12   0.001                             X
 13   0.161                             XXXXX
 14  -0.021                           XX
 15   0.053                             XX
 16  -0.090                          XXX
 17  -0.136                          XXXX
 18  -0.044                            XX
 19  -0.066                          XXX
 20  -0.078                          XXX
```

Cut off

ARIMA Model: unemploy

```
Modified Box-Pierce (Ljung-Box) Chi-Square statistic
Lag                 12         24         36         48
Chi-Square        17.9       32.8       44.1       50.1
DF                   9         21         33         45
P-Value          0.036      0.048      0.094      0.278
```

```
Series: unemploy
ARIMA(2,0,0) with non-zero mean

Coefficients:
         ar1      ar2    mean
      1.5476  -0.6450  5.0819
s.e.  0.0682   0.0687  0.3275

sigma^2 estimated as 0.1318:  log likelihood=-49.17
AIC=106.34   AICc=106.68   BIC=117.52
```

ARIMA Model: unemploy

```
Modified Box-Pierce (Ljung-Box) Chi-Square statistic
Lag                12        24        36        48
Chi-Square       15.7      31.2      40.6      46.7
DF                  8        20        32        44
P-Value         0.047     0.052     0.142     0.362
```

```
Series: unemploy
ARIMA(2,0,1) with non-zero mean

Coefficients:
         ar1      ar2     ma1    mean
      1.4194  -0.5245  0.2286  5.0834
s.e.  0.1133   0.1109  0.1291  0.3651

sigma^2 estimated as 0.1295:  log likelihood=-47.65
AIC=105.29   AICc=105.82   BIC=119.27
```

ARMA(2,1)

Hence AIC and AICC choose ARMA(2,1) model, while for BIC, the model is AR(2).

Example: Calculation of Box-Pierce statistic and Ljung-Box statistics

```
Modified Box-Pierce (Ljung-Box) Chi-Square statistic
Lag                12        24        36        48
Chi-Square       17.9      32.8      44.1      50.1
DF                  9        21        33        45
P-Value         0.036     0.048     0.094       0.2
```

error terms of AR(2) "ACF"

ACF of ar2-err

```
          -1.0 -0.8 -0.6 -0.4 -0.2  0.0  0.2  0.4  0.6  0.8  1.0
            +----+----+----+----+----+----+----+----+----+----+
  1   0.103                          XXXX
  2  -0.056                        XX
  3  -0.037                        XX
  4  -0.112                      XXXX
  5  -0.040                        XX
  6   0.133                         XXXX
  7   0.190                         XXXXXX
  8  -0.136                      XXXX
  9   0.072                         XXX
 10   0.091                         XXX
 11   0.111                         XXXX
 12  -0.095                       XXX
```

Hence the Box-Pierce statistic and Ljung-Box statistic are given by

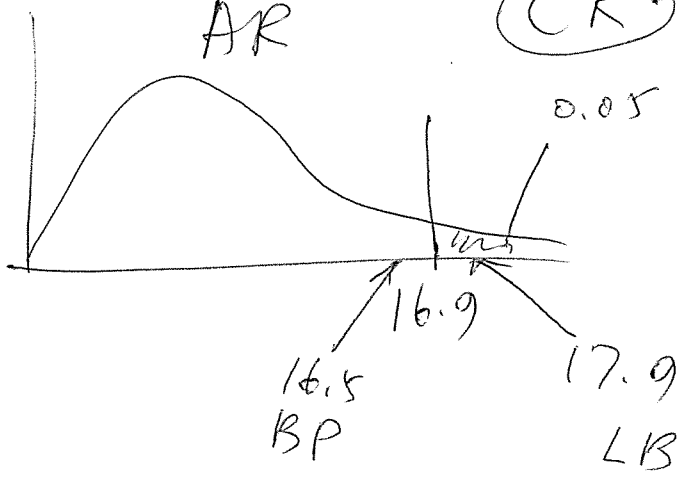
BP(up to twelve terms)=$121(0.103^2 + (-0.056)^2 + \ldots + 0.111^2 + (-0.905)^2) = 16.5$

LB(up to twelve terms)=$121(123)[0.103^2/120 + (-0.056)^2/119 + \ldots + (-0.095)^2/109]$
$= 17.9$

$n - k = 121 - 12$
$= 109$

As $\chi^2_{9,\,0.05} = 16.9$, the BP statistic is not significant at 5% level but LB is.

do not reject Ho = Acceptance Region



Critical Region

reject Ho